

ECE 269 Discussion 4

1 Left Inverse and Right Inverse

Consider $\mathbf{A} = [\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n] \in \mathbb{F}^{m \times n}$, where $\mathbf{a}_i \in \mathbb{F}^m$. From the lecture, we have already known $\mathcal{R}(\mathbf{A}) = \text{span}\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n\}$ and $\mathcal{R}(\mathbf{A})$ is a subspace of $\mathbb{F}^m$.

1.1 Right inverse

Given $\mathbf{A} \in \mathbb{F}^{m \times n}$, there exists $\mathbf{B} = [\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_m] \in \mathbb{F}^{n \times m}$ such that $\mathbf{AB} = \mathbf{I}$ if and only if $\text{rank}(\mathbf{A}) = m$. Such a matrix $\mathbf{B}$ will be called a right inverse of $\mathbf{A}$.

Proof:

Claim: $\mathbf{AB} = \mathbf{I} = [\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_m] \Leftrightarrow \{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_m\} \in \mathcal{R}(\mathbf{A})$.

(Why? If $\mathbf{AB} = \mathbf{I}$, we have $\mathbf{e}_i = \mathbf{Ab}_i$, which means $\mathbf{e}_i \in \mathcal{R}(\mathbf{A})$ for $i = 1, 2, \dots, m$. Conversely, if $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_m \in \mathcal{R}(\mathbf{A})$, then there exists $\mathbf{b}_i$ such that $\mathbf{Ab}_i = \mathbf{e}_i$.

Now, let us prove that $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_m\} \in \mathcal{R}(\mathbf{A}) \Leftrightarrow \dim(\mathcal{R}(\mathbf{A})) = m$.

First, suppose $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_m\} \in \mathcal{R}(\mathbf{A})$, we know that $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_m\}$ is a basis of $\mathbb{F}^m$ and $\mathcal{R}(\mathbf{A})$ is a subspace of $\mathbb{F}^m$. Therefore, $\mathcal{R}(\mathbf{A}) \subseteq \text{span}\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_m\}$. Since $\mathbf{e}_1, \dots, \mathbf{e}_m \in \mathcal{R}(\mathbf{A})$, then we also have $\text{span}\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_m\} \subseteq \mathcal{R}(\mathbf{A})$, implying that $\text{span}\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_m\} = \mathcal{R}(\mathbf{A})$. Since $\mathbf{e}_1, \dots, \mathbf{e}_m$ are linear independent, they are a basis of $\mathcal{R}(\mathbf{A})$, which means $\dim(\mathcal{R}(\mathbf{A})) = m$.

Conversely, suppose $\dim(\mathcal{R}(\mathbf{A})) = m$, since $\mathcal{R}(\mathbf{A})$ is a subspace of $\mathbb{F}^m$, we have $\mathcal{R}(\mathbf{A}) = \mathbb{F}^m$. Therefore, $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_m\} \in \mathcal{R}(\mathbf{A})$.

1.2 Left inverse

Given $\mathbf{A} \in \mathbb{F}^{m \times n}$, there exists $\mathbf{C} \in \mathbb{F}^{n \times m}$ such that $\mathbf{CA} = \mathbf{I}$ if and only if $\text{rank}(\mathbf{A}) = n$. Such a matrix $\mathbf{C}$ will be called a left inverse of $\mathbf{A}$.

Proof:

DIY. (Hint: Apply the proof of right inverse on $\mathbf{A}^\top$ and use the fact that $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A}^\top)$.)

2 The inverse of a matrix A

Q: Can a matrix $\mathbf{A} \in \mathbb{F}^{m \times n}$ have both right and left inverse?

Suppose $\mathbf{A}$ has both left inverse $\mathbf{C}$ and right inverse $\mathbf{B}$, we have $(\mathbf{CA})\mathbf{B} = \mathbf{IB} = \mathbf{B}$ and $\mathbf{CAB} = \mathbf{CI} = \mathbf{C}$, therefore $\mathbf{B} = \mathbf{C}$.

Q: Is $m = n$?

Suppose $n \neq m$, then either $n > m$ or $m > n$. If $n > m$, then $\text{rank}(\mathbf{A}) < n$, which means no left inverse of $\mathbf{A}$ exists. Similarly, if $m > n$, then $\text{rank}(\mathbf{A}) < m$, which means no right inverse of $\mathbf{A}$ exists.

In conclusion, if $\mathbf{A}$ has both left and right inverse, the left and right inverse must be the same matrix and

we must have $\text{rank}(\mathbf{A}) = m = n$.

Such an inverse of $\mathbf{A}$, which is both left and reverse of $\mathbf{A}$, is called the inverse of $\mathbf{A}$. We call such a matrix with $\text{rank}(\mathbf{A}) = m = n$ as an **invertible matrix**. The inverse of $\mathbf{A}$ is often denoted as $\mathbf{A}^{-1}$.

$\mathbf{A}^{-1} \in \mathbb{F}^{n \times n}$ is a matrix such that $\mathbf{A}\mathbf{A}^{-1} = \mathbf{I} = \mathbf{A}^{-1}\mathbf{A}$, and there is a unique matrix $\mathbf{A}^{-1}$ which satisfies the above.

Now, given $\mathbf{A} \in \mathbb{F}^{m \times n}$, consider the following three cases:

- (1) $\text{rank}(\mathbf{A}) = m \Rightarrow$ right inverse of $\mathbf{A}$ exists $\Rightarrow$ right inverse is non-unique if $n > m$ ($\mathbb{F}$ not finite), is unique if $n = m$.
- (2) $\text{rank}(\mathbf{A}) = n \Rightarrow$ left inverse of $\mathbf{A}$ exists $\Rightarrow$ left inverse is non-unique if $m > n$ ($\mathbb{F}$ not finite), is unique if $m = n$.
- (3) $\text{rank}(\mathbf{A}) = m = n \Rightarrow \mathbf{A}$ is invertible $\Rightarrow$ inverse is unique.

3 Problem 1

Given S_1 and S_2 are subspaces, then

$$\dim(S_1 + S_2) = \dim(S_1) + \dim(S_2) - \dim(S_1 \cap S_2),$$

where $S_1 + S_2 = \{\mathbf{x}_1 + \mathbf{x}_2 | \mathbf{x}_1 \in S_1, \mathbf{x}_2 \in S_2\}$.

Proof:

Suppose $\dim(S_1) = k_1$, $\dim(S_2) = k_2$, $\dim(S_1 \cap S_2) = k$ and the basis of $S_1 \cap S_2$ be $\{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_k\}$. As all the basis vectors given above for $S_1 \cap S_2$ belong to both S_1 and S_2 , we can extend a basis for S_1 and S_2 : let the basis for S_1 be $\{\mathbf{b}_1, \dots, \mathbf{b}_k, \mathbf{b}_{1,k+1}, \dots, \mathbf{b}_{1,k_1}\}$ (k_1 independent vectors), let the basis for S_2 be $\{\mathbf{b}_1, \dots, \mathbf{b}_k, \mathbf{b}_{2,k+1}, \dots, \mathbf{b}_{2,k_2}\}$ (k_2 independent vectors).

- $S_1 + S_2$ should contain all the vectors in S_1 and S_2 :

$$\text{span}\{\mathbf{b}_1, \dots, \mathbf{b}_k, \mathbf{b}_{1,k+1}, \dots, \mathbf{b}_{1,k_1}, \mathbf{b}_{2,k+1}, \dots, \mathbf{b}_{2,k_2}\} \subseteq S_1 + S_2$$

- $\forall \mathbf{x}_1 \in S_1$ and $\forall \mathbf{x}_2 \in S_2$, we have

$$\mathbf{x}_1 = \alpha_1 \mathbf{b}_1 + \alpha_2 \mathbf{b}_2 + \dots + \alpha_{k_1} \mathbf{b}_{1,k_1}, \quad \mathbf{x}_2 = \beta_1 \mathbf{b}_1 + \beta_2 \mathbf{b}_2 + \dots + \beta_{k_2} \mathbf{b}_{2,k_2}.$$

Hence every vector in $S_1 + S_2$ can be written as $\mathbf{x}_1 + \mathbf{x}_2 = \gamma_1 \mathbf{b}_1 + \gamma_2 \mathbf{b}_2 + \dots + \gamma_{1,k_1} \mathbf{b}_{1,k_1} + \dots + \gamma_{2,k_2} \mathbf{b}_{2,k_2}$, which implies $S_1 + S_2 \subseteq \text{span}\{\mathbf{b}_1, \dots, \mathbf{b}_k, \mathbf{b}_{1,k+1}, \dots, \mathbf{b}_{1,k_1}, \mathbf{b}_{2,k+1}, \dots, \mathbf{b}_{2,k_2}\}$.

- Now let us verify whether $\{\mathbf{b}_1, \dots, \mathbf{b}_k, \mathbf{b}_{1,k+1}, \dots, \mathbf{b}_{1,k_1}, \mathbf{b}_{2,k+1}, \dots, \mathbf{b}_{2,k_2}\}$ are linear independent. Suppose there exists non-zero solution for

$$\alpha_1 \mathbf{b}_1 + \alpha_2 \mathbf{b}_2 + \dots + \alpha_{1,k_1} \mathbf{b}_{1,k_1} + \alpha_{2,k+1} \mathbf{b}_{2,k+1} + \dots + \alpha_{2,k_2} \mathbf{b}_{2,k_2} = \mathbf{0}$$

Rewriting the above equation, we get

$$\alpha_1 \mathbf{b}_1 + \alpha_2 \mathbf{b}_2 + \dots + \alpha_{1,k_1} \mathbf{b}_{1,k_1} = \alpha'_{2,k+1} \mathbf{b}_{2,k+1} + \dots + \alpha'_{2,k_2} \mathbf{b}_{2,k_2} = \mathbf{s}, \quad (1)$$

where α' denotes additive inverse of α . Here the first term in (1) $\in S_1$ and the second term in (1) $\in S_2$, which means $\mathbf{s} \in S_1 \cap S_2$. Since any vector in $S_1 \cap S_2$ can be written as $\mathbf{s} = \gamma_1 \mathbf{b}_1 + \gamma_2 \mathbf{b}_2 + \dots + \gamma_k \mathbf{b}_k$, substituting this into (1) we have

$$\alpha_1 \mathbf{b}_1 + \alpha_2 \mathbf{b}_2 + \dots + \alpha_{1,k_1} \mathbf{b}_{1,k_1} = \gamma_1 \mathbf{b}_1 + \gamma_2 \mathbf{b}_2 + \dots + \gamma_k \mathbf{b}_k$$

$$(\alpha_1 - \gamma_1)\mathbf{b}_1 + (\alpha_2 - \gamma_2)\mathbf{b}_2 + \cdots + (\alpha_k - \gamma_k)\mathbf{b}_k + \alpha_{1,k+1}\mathbf{b}_{1,k+1} + \cdots + \alpha_{1,k_1}\mathbf{b}_{1,k_1} = \mathbf{0}$$

Since all these vectors form a basis for S_1 , they must be linear independent. Hence, $\alpha_{1,k+1} = \cdots = \alpha_{1,k_1} = 0$. Substituting this into (1), we get

$$\alpha_1\mathbf{b}_1 + \alpha_2\mathbf{b}_2 + \cdots + \alpha_k\mathbf{b}_k = \alpha'_{2,k+1}\mathbf{b}_{2,k+1} + \cdots + \alpha'_{2,k_2}\mathbf{b}_{2,k_2}$$

$$\alpha_1\mathbf{b}_1 + \alpha_2\mathbf{b}_2 + \cdots + \alpha_k\mathbf{b}_k + \alpha_{2,k+1}\mathbf{b}_{2,k+1} + \cdots + \alpha_{2,k_2}\mathbf{b}_{2,k_2} = \mathbf{0}.$$

As these vectors form a basis for S_2 , they must be linear independent. Therefore, $\alpha_1 = \cdots = \alpha_k = 0$ and $\alpha_{2,k+1} = \cdots = \alpha_{2,k_2} = 0$. As all coefficients are 0, it contradicts the assumption and concludes that $\{\mathbf{b}_1, \cdots, \mathbf{b}_k, \mathbf{b}_{1,k+1}, \cdots, \mathbf{b}_{1,k_1}, \mathbf{b}_{2,k+1}, \cdots, \mathbf{b}_{2,k_2}\}$ are linear independent.

Thus, $\{\mathbf{b}_1, \cdots, \mathbf{b}_k, \mathbf{b}_{1,k+1}, \cdots, \mathbf{b}_{1,k_1}, \mathbf{b}_{2,k+1}, \cdots, \mathbf{b}_{2,k_2}\}$ forms a basis for $S_1 + S_2$.

$$\dim(S_1 + S_2) = k + k_1 - k + k_2 - k = k_1 + k_2 - k = \dim(S_1) + \dim(S_2) - \dim(S_1 \cap S_2).$$